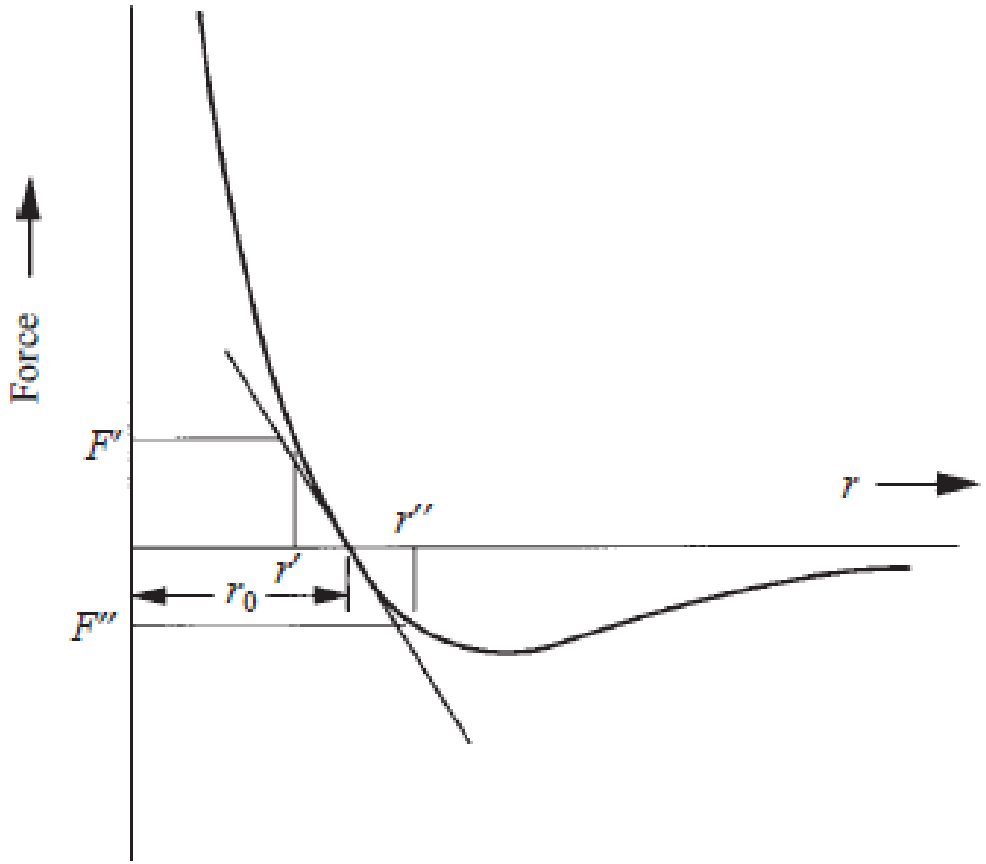
Mechanical Properties (Part 2)

MO-201 Materials Science
Odd Semester 2026-27
Dr. S. Sinha (Met. Engg.)

References:
Materials Science & Engg. by V. Raghavan
Fundamentals of Materials Science by W.D. Callister

Atomic model of elastic behaviour



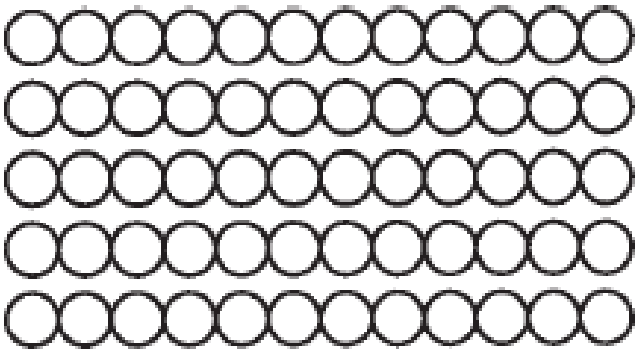
$$W = -\frac{A}{r^n} + \frac{B}{r^m}$$

$$F = -\frac{dW}{dr}$$

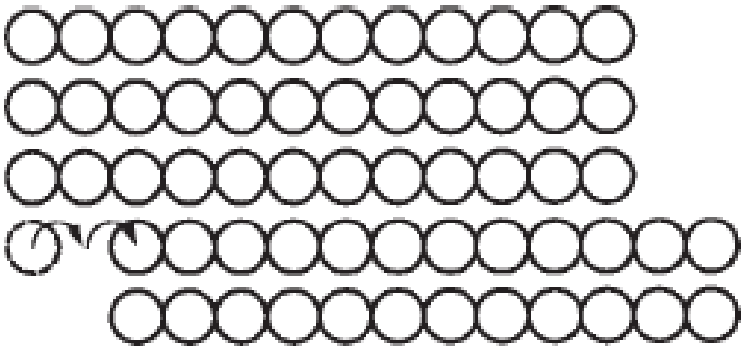
$$Y \propto -\frac{dF}{dr} = \frac{d^2W}{dr^2}$$

The interatomic force-distance curve. Changing the bond length from r_0 to r' or r'' requires an external force of magnitude F' or F''.

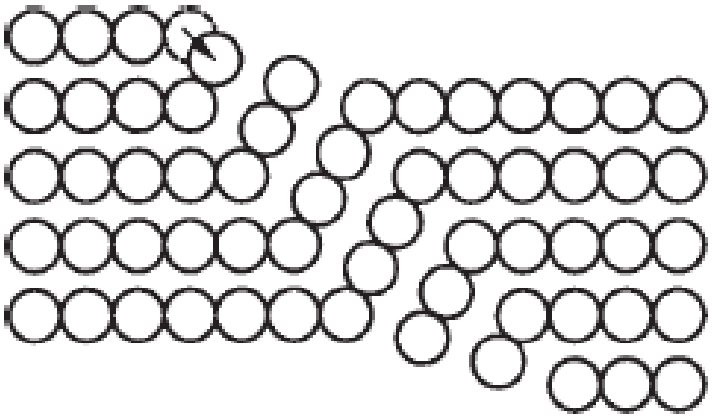
Deformation by slip and twinning



(a)
Before slip or twinning



(b)
After slip



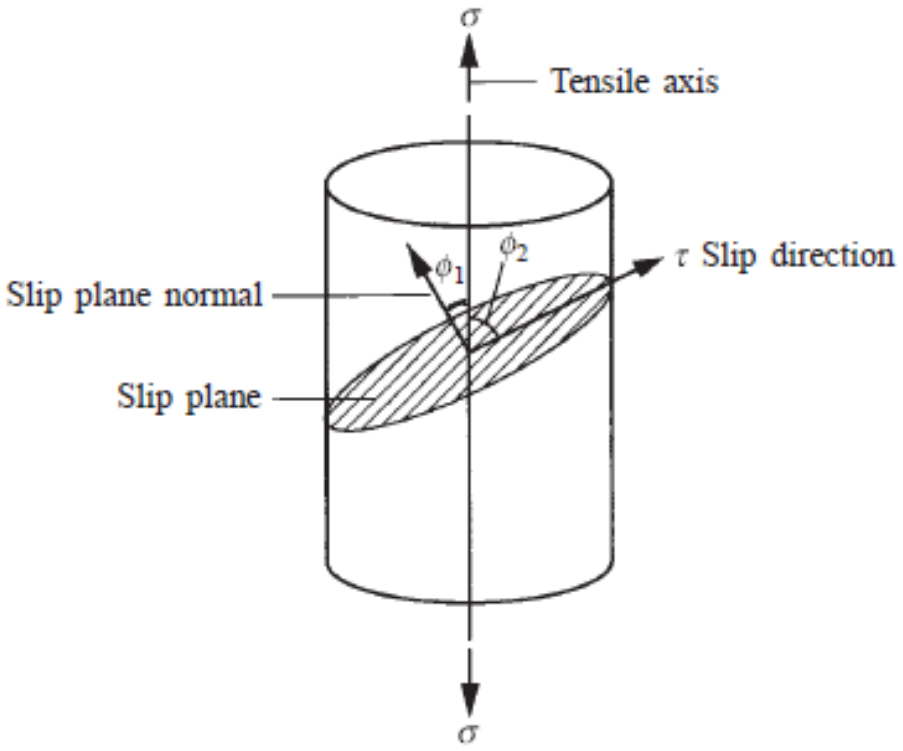
(c)
After twinning

Slip systems

Slip Planes and Directions in Some Crystals

Crystal		Slip planes	Slip directions
FCC		{111}	⟨110⟩
BCC	more common	{110}	⟨111⟩
	less common	{112}, {123}	
NaCl		{110}	⟨110⟩
HCP	more common	basal plane	close packed
	less common	prismatic and pyramidal planes	directions

CRSS and Schmid Factor



$$\tau = \sigma \cos \phi_1 \cos \phi_2$$

$\cos \phi_1 \cos \phi_2$ = Schmid Factor

This resolved shear stress should reach a critical value called the *critical resolved shear stress* (CRSS) for plastic deformation to start. It is evident that all slip systems in a crystal will not have the same resolved shear stress for a given tensile stress along an axis. As the applied tensile stress is increased from zero, deformation will be initiated *first* on that slip system for which *the resolved shear stress is a maximum* and so reaches the critical value first.

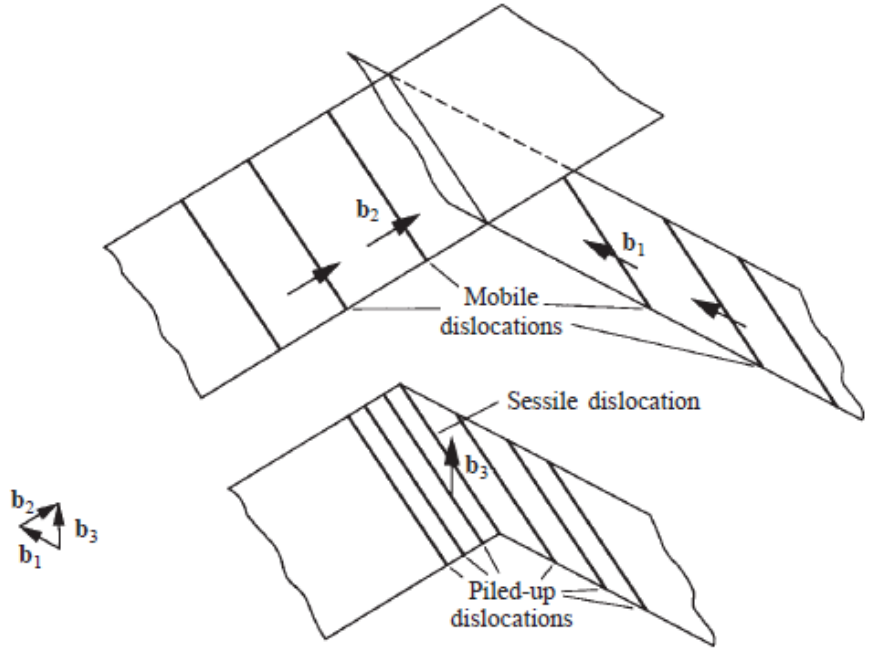
Dislocation reaction

Formation of a sessile dislocation in an FCC crystal by a dislocation reaction.

For example, consider the dislocations moving on nonparallel planes of an FCC crystal. The slip planes in FCC belong to the {111} family. Let a dislocation of Burgers vector $1/2[0\,\bar{1}\,1]$ moving on $(\bar{1}\,11)$ plane meet another dislocation of Burgers vector $1/2[10\,\bar{1}]$ moving on $(1\,\bar{1}\,1)$ plane along the line of intersection of these two planes (see Figure) These two dislocations can interact and produce a third dislocation with a decrease in the potential energy:

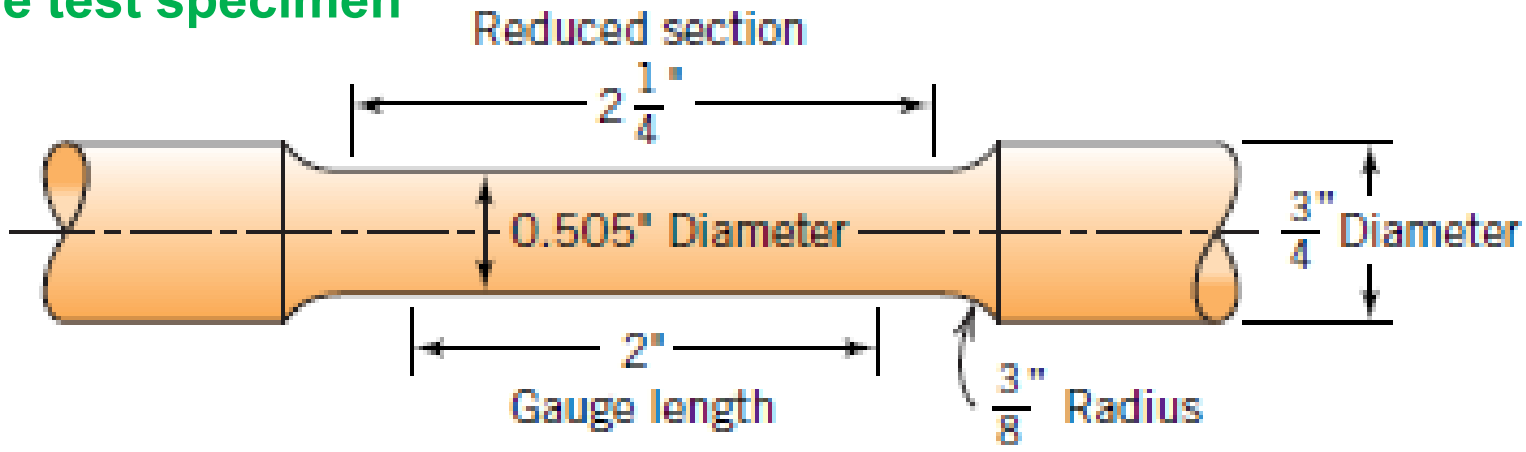
$$\frac{1}{2}[0\,\bar{1}\,1] + \frac{1}{2}[10\,\bar{1}] \rightarrow \frac{1}{2}[1\,\bar{1}\,0]$$

The strain energy decreases by 50% during this dislocation reaction. As the Burgers vector of the product dislocation lies in neither of these two slip planes, it becomes immobile (or sessile)



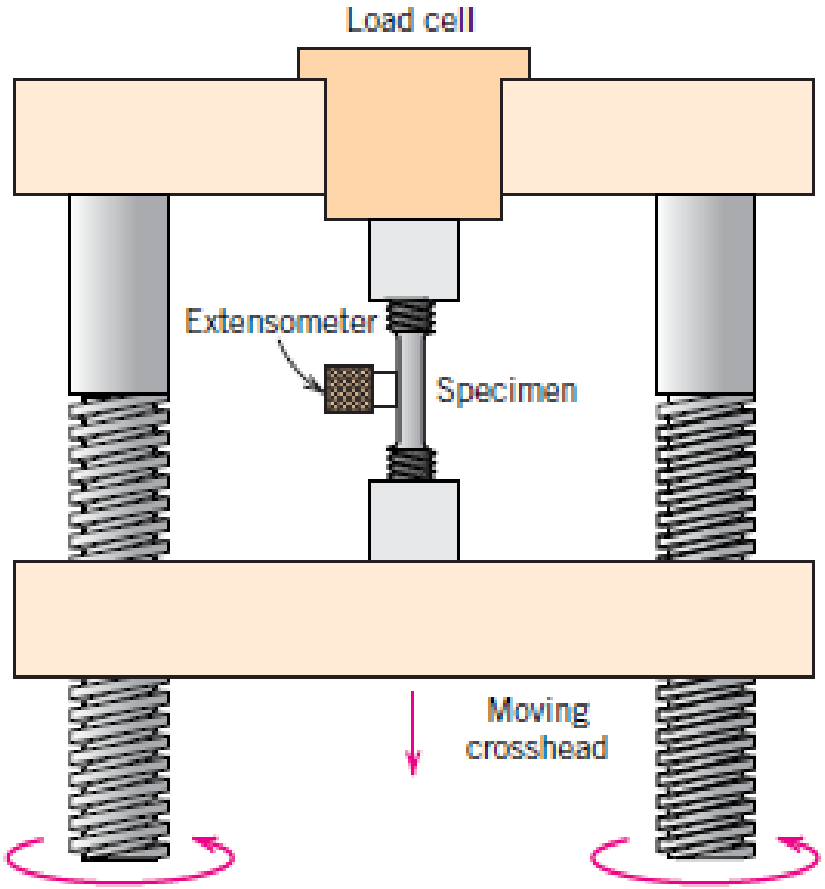
Tensile Test

Tensile test specimen



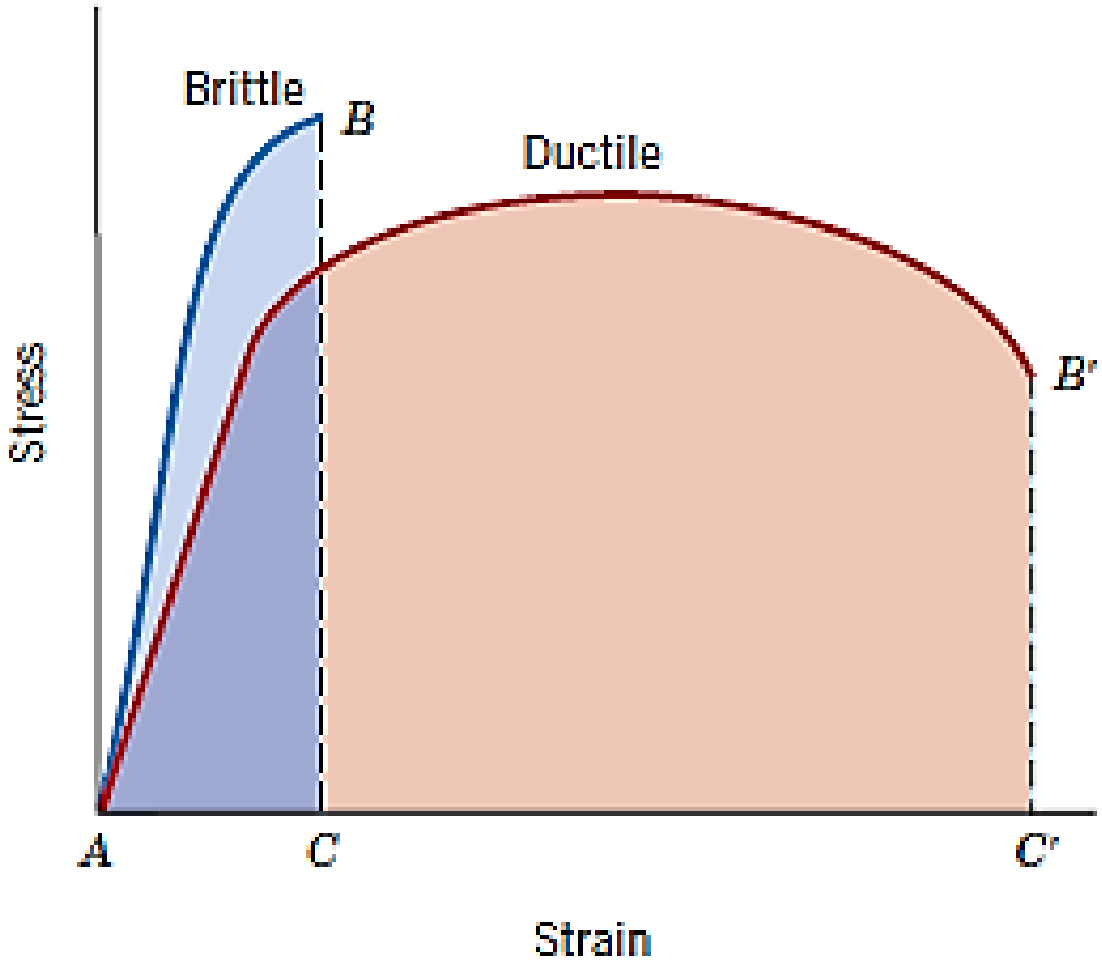
- Load
- Elongation/Displacement
- Stress
- Strain

Tensile Test



Universal Testing Machine (UTM)

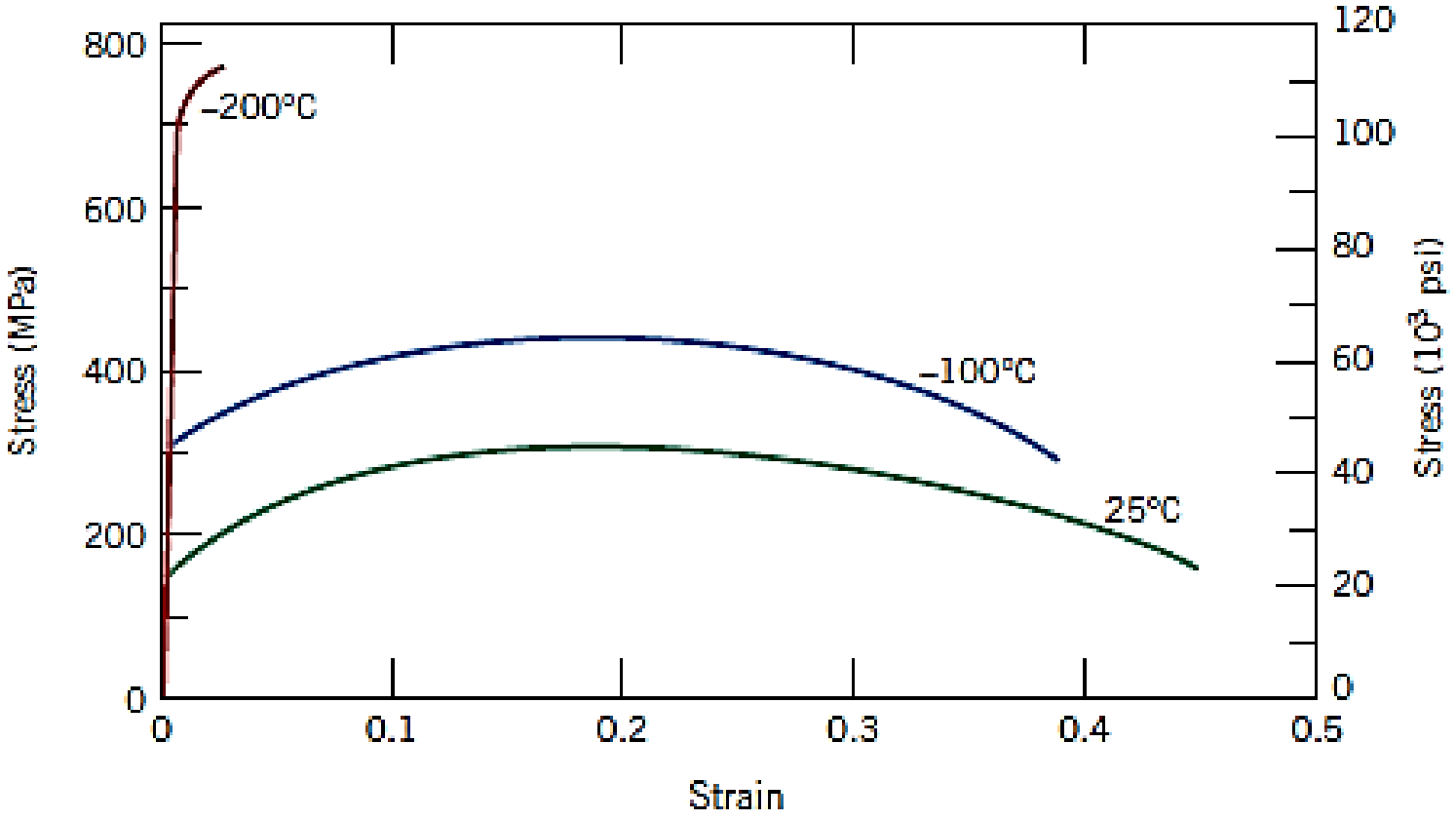
Tensile Test



Tensile stress-strain curve

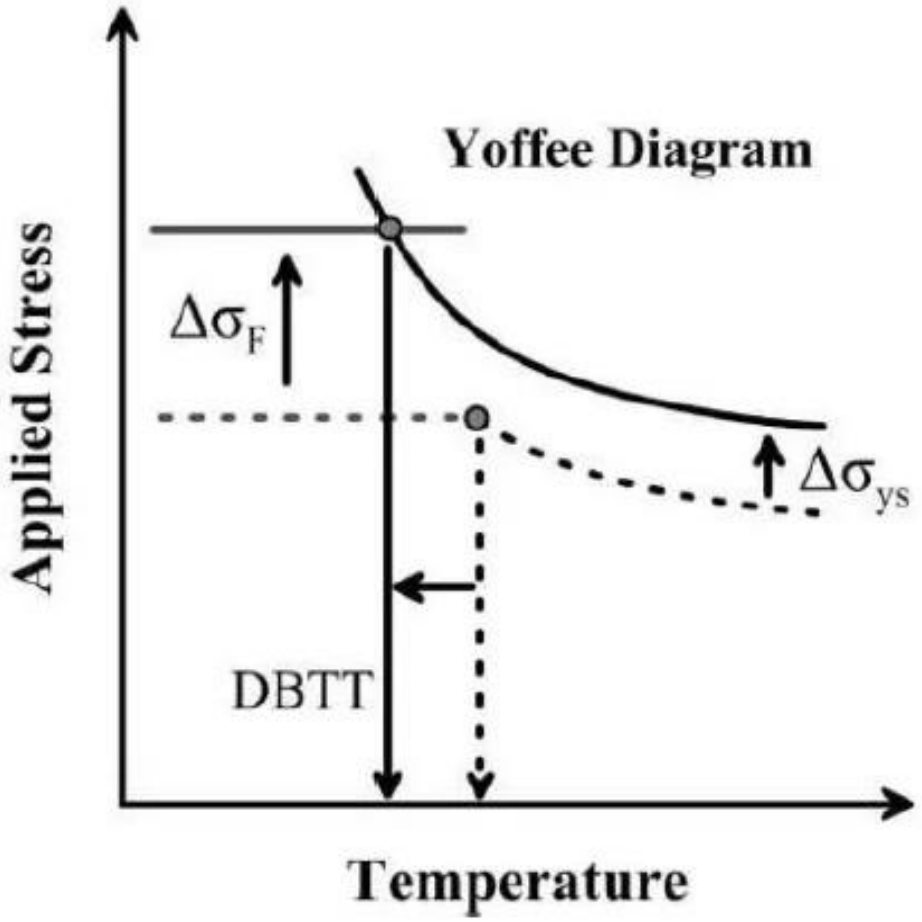
Tensile Test

Brittle and Ductile



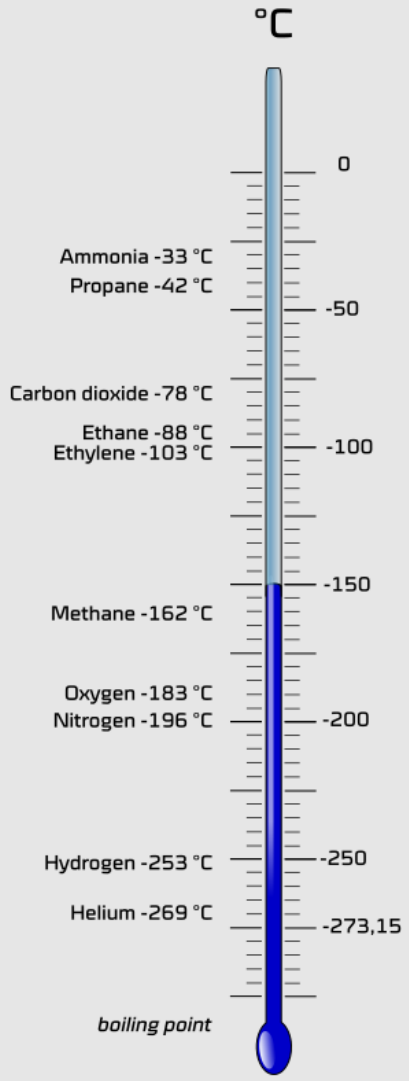
For example, all 3 curves here are for iron

Yoffee Diagram



https://knifesteelnerds.com/2018/12/21/why-cold-steel-is-brittle/

Materials resistant to extremely low T



STEELS	ALUMINUM ALLOYS	NICKEL ALLOYS	TITANIUM ALLOYS
Ultra-High Strength Steels Carbon steels	7178-T6		Heat treated 13V-11Cr-3Al-Ti Heat treated 8Mn-Ti
Fine grained carbon steels	7075-T6		
			Heat treated 6Al-6V-2Sn-Ti
Heat-treated alloy steels T-1, N-A-XTRA, AMS6434, 4335, etc.			
HY-TUF, HY-80 3.5% Ni steel			
			Heat treated 16V-2.5Al-Ti
5% Ni steel	7079-T6		
			8Al-1Mo-1V-Ti, 5Al-5Zr-5Sn-Ti Annealed 4Al-3Mo-1V-Ti
			Heat treated 6Al-4V-Ti Heat treated 8Al-2Cb-1Ta-Ti
Quenched and tempered 9% Ni steel A286, Maraging steels 301, 302	2024-T6, 7039-T6 2014-T6, 5456-H343		
			Annealed 6Al-4V-Ti ELI
304ELC, 310 Low-C stainless steel casting alloys	2024-T4, 6061-T6 2219-T87, 5052-H38 5083-H38	Monel, TD Nickel Hastelloy B, Inconel X Inconel 718, René 41	5Al-2.5Sn-Ti ELI, Ti45A(AMS 4902)

https://www.gasparini.com/en/blog/metals-and-materials-for-low-temperatures/

Tensile Test

Properties from tensile test

- Elastic modulus and Resilience
- Yield Stress (YS)
- Tensile Strength (TS) or Ultimate Tensile Strength (UTS)
- Elongation to failure/Reduction of area at fracture
- Toughness

Tensile Test

Anelasticity

- In most engineering materials, there is a time-dependent elastic strain component (elastic deformation will continue after the stress application, and upon load release, some finite time is required for complete recovery)

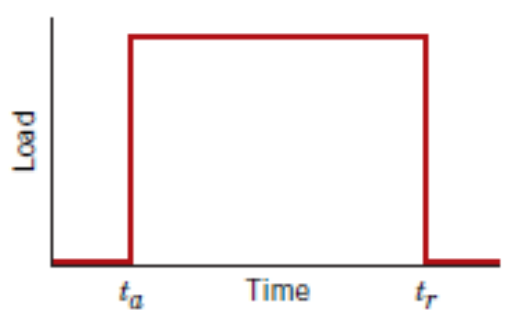
- This time-dependent elastic behavior is known as *anelasticity*, and it is due to time-dependent microscopic and atomistic processes that are attendant to the deformation

- For metals, the anelastic component is normally small and is often neglected; however, for some polymeric materials, its magnitude is significant – in this case it is termed viscoelastic behavior (rubber-like)

Tensile Test

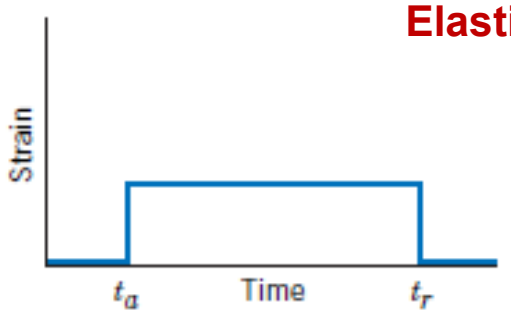
Anelasticity

Load-time cycle

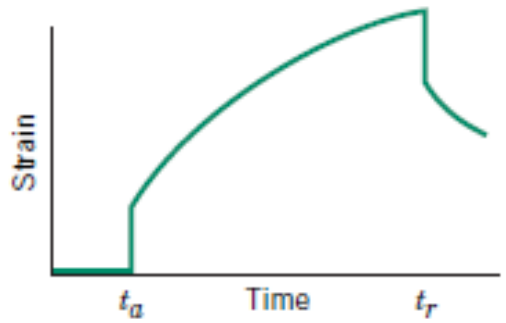


(a)

Elastic

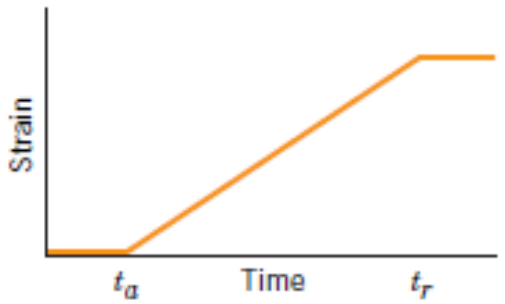


(b)



(c)

Viscoelastic



(d)

Viscous

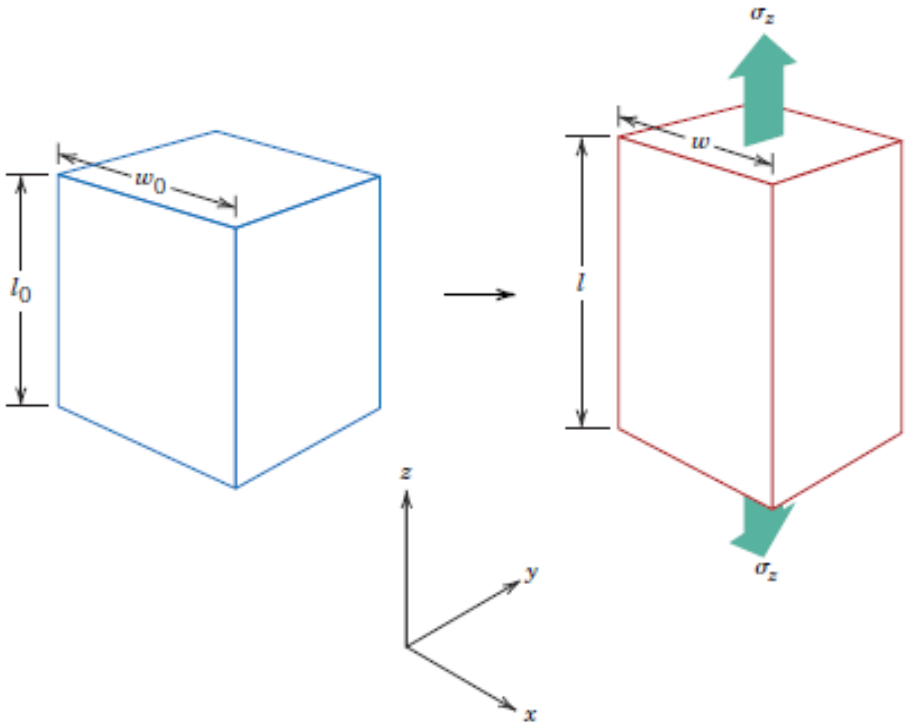
Tensile Test

Poisson's Ratio

- When a tensile stress is imposed on a metal specimen, an elastic elongation and accompanying strain ε_z result in the direction of the applied stress (arbitrarily taken to be the z direction)

- As a result of this elongation, there will be constrictions in the lateral (x and y) directions perpendicular to the applied stress (compressive strains ε_x and ε_y)

- ν typically 0.25-0.35 for metals/alloys
  0.33-0.46 for polymers

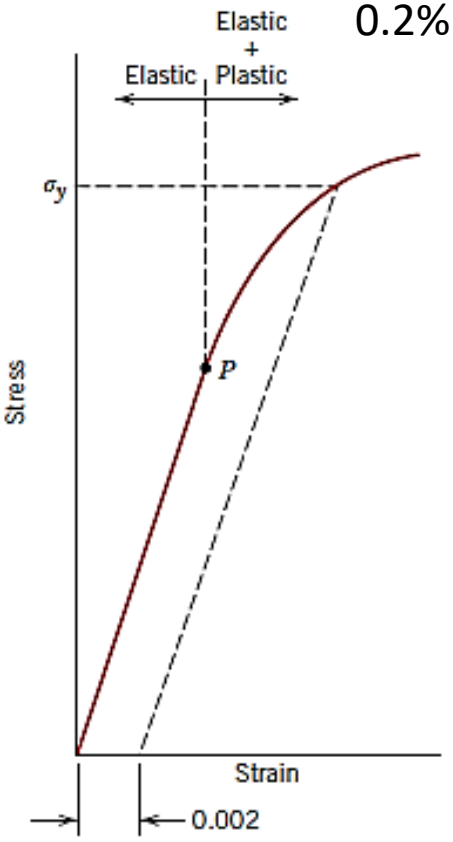


$$\nu = -\frac{\varepsilon_x}{\varepsilon_z} = -\frac{\varepsilon_y}{\varepsilon_z}$$ (ε_x=ε_y for isotropic)

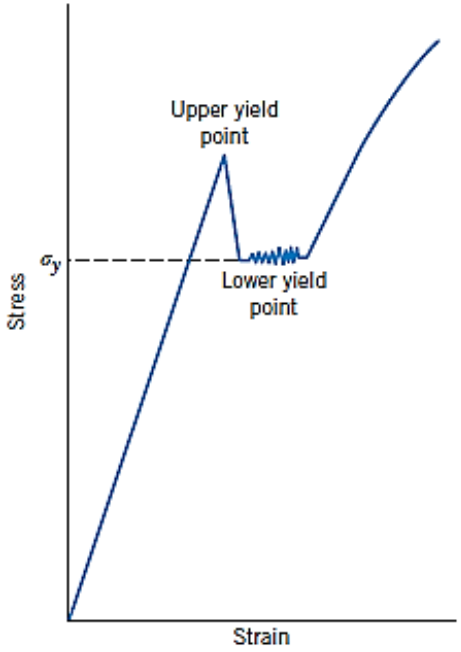
$$E = 2G(1+\nu)$$

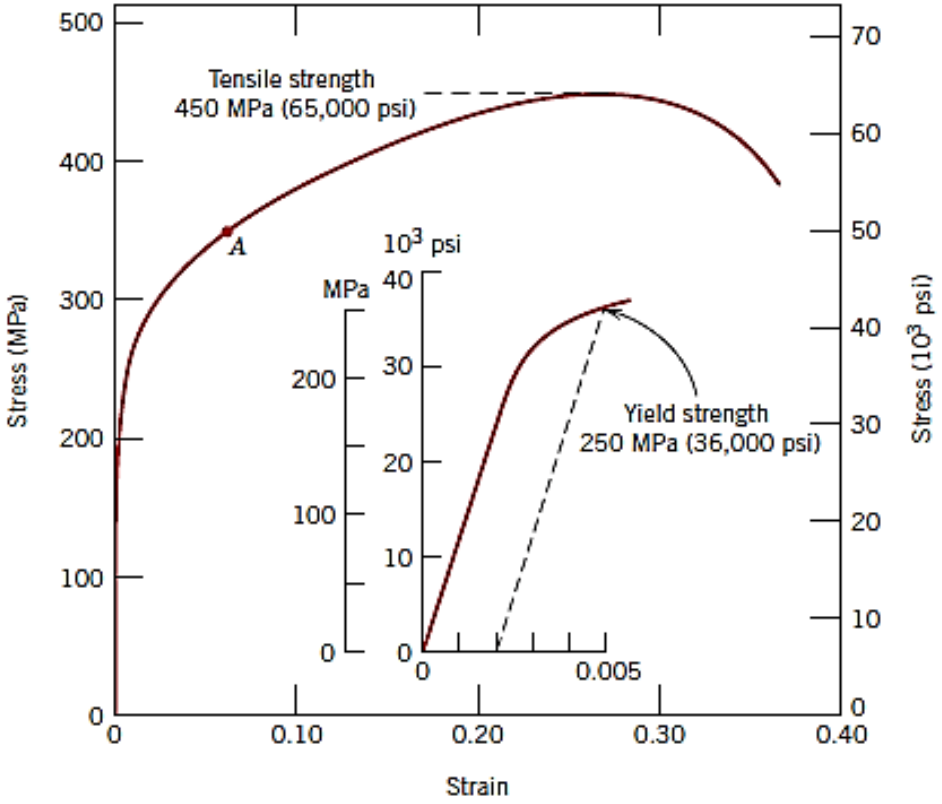
Tensile Test

Yield Stress and Tensile Strength



0.2% offset Yield Stress





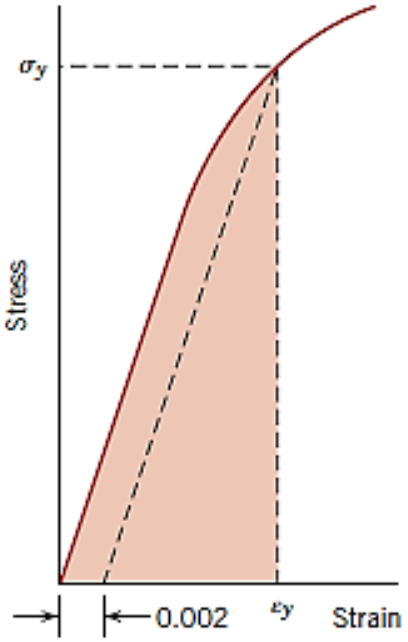
Regime of plastic deformation

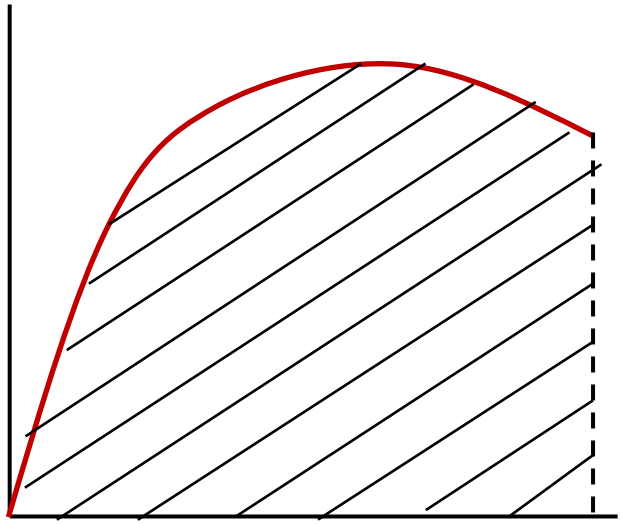
Tensile Test

Resilience and Toughness

Resilience

Area under curve up to yield





Toughness

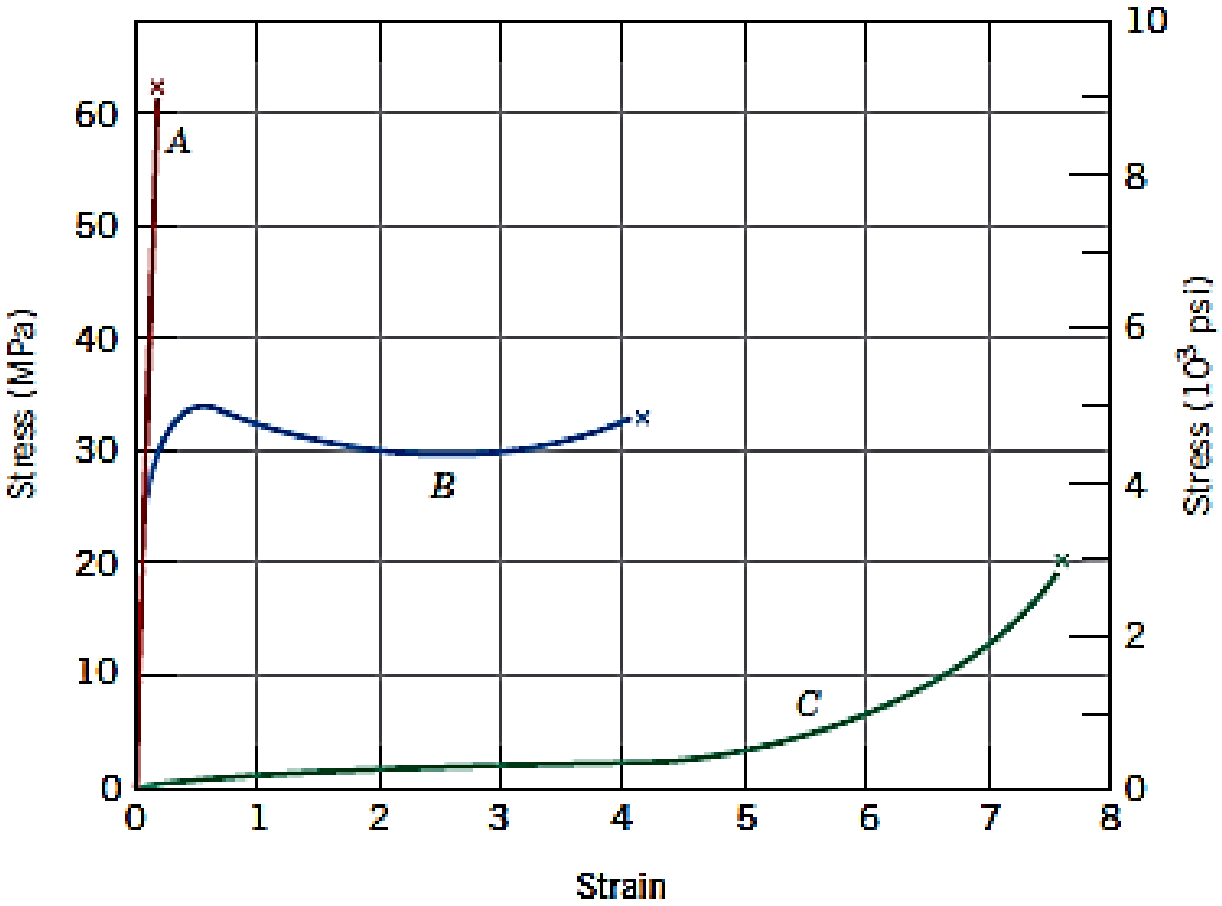
Area under curve till fracture

Modulus of resilience

$$U_r = \int_0^{\varepsilon_y} \sigma d\varepsilon = \frac{1}{2}\sigma_y \varepsilon_y = \frac{{\sigma_y}^2}{2E}$$

Tensile Test

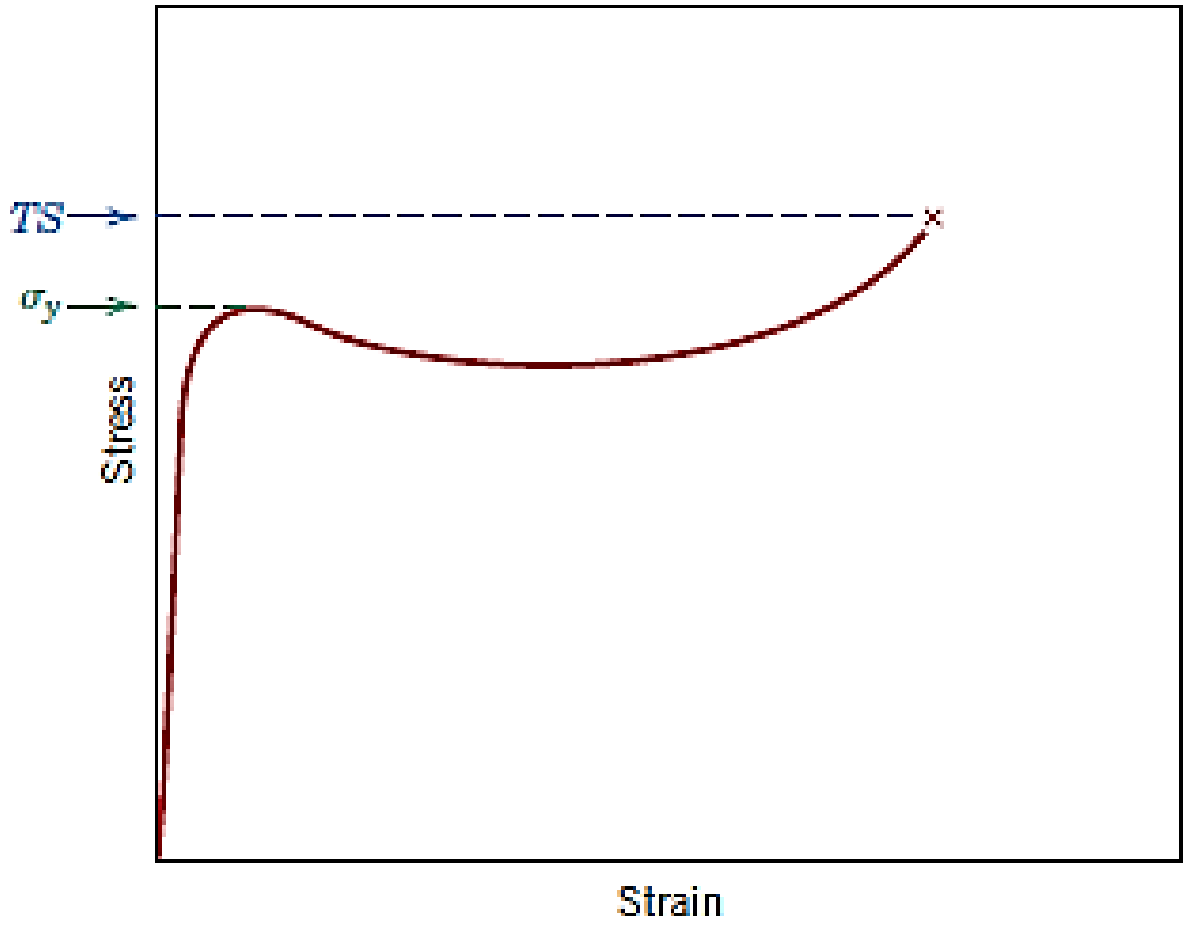
Tensile curves for polymers



- Brittle
- Plastic
- Elastomeric (highly elastic)

Tensile Test

Tensile curves for polymers



Tensile Test

Measures of strength and ductility

Engineering Stress

$$\sigma = \frac{F}{A_0}$$

Engineering Strain

$$\varepsilon = \frac{l_i - l_0}{l_0}$$

Percent Elongation

$$\%EL = \left(\frac{l_f - l_0}{l_0}\right) \times 100$$

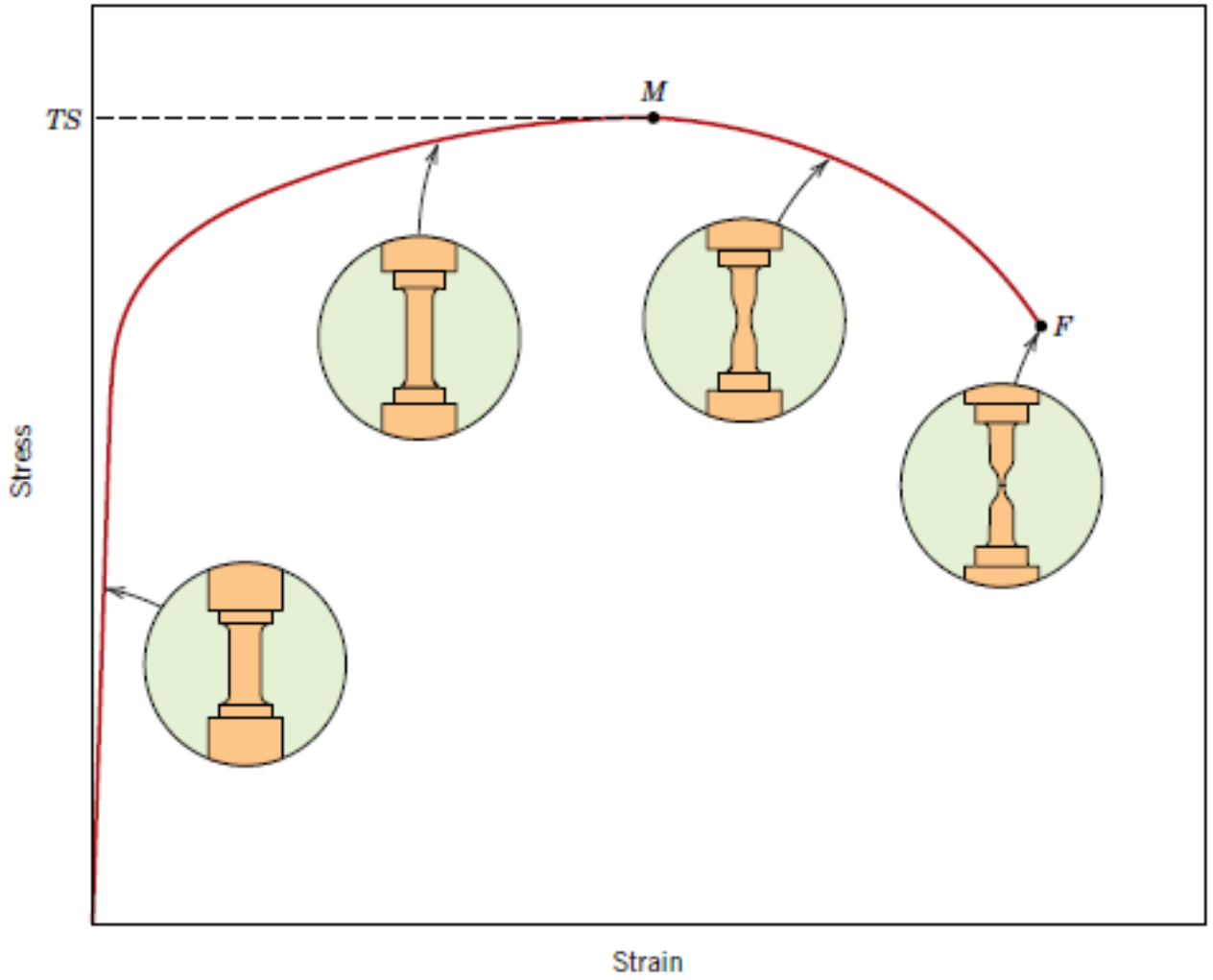
Percent Reduction in Area

$$\%RA = \left(\frac{A_0 - A_f}{A_0}\right) \times 100$$

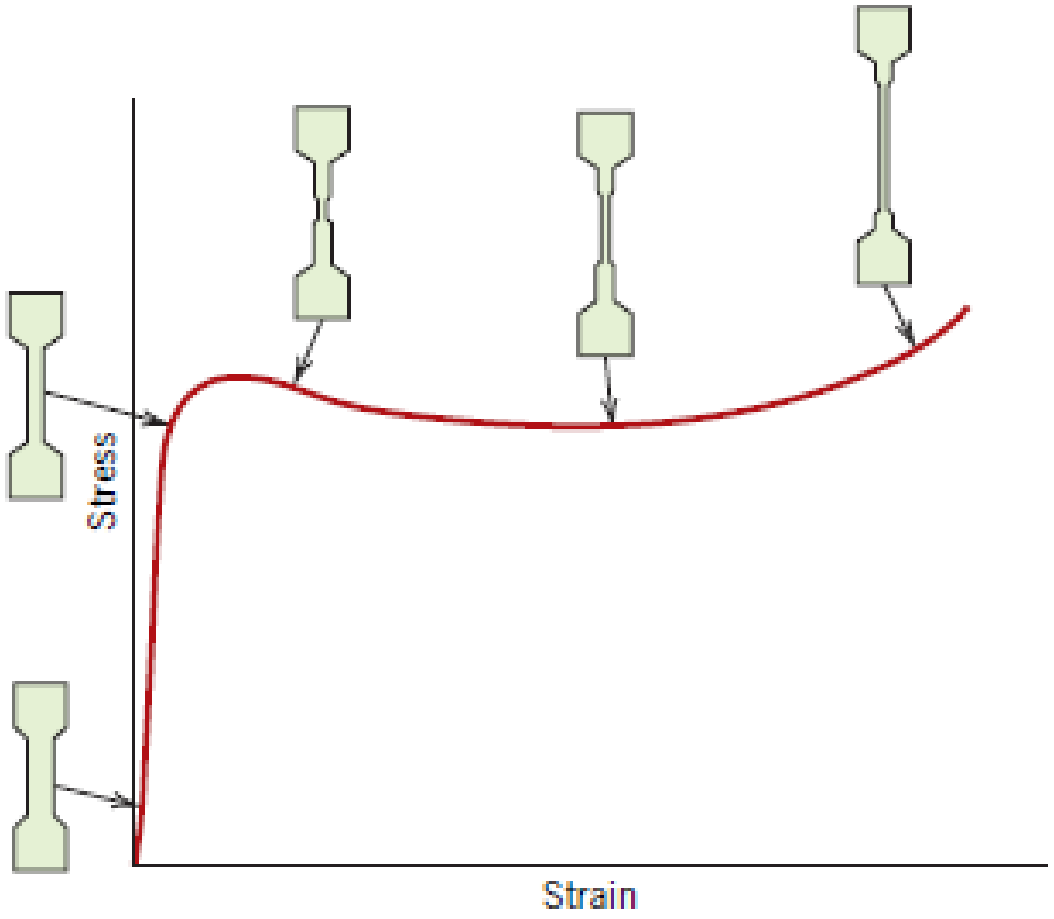
- Yield strength (YS) and Tensile Strength (TS)/UTS are measures of strength
- Fracture strength is not used as a design criterion
- Percent Elongation to Failure (%EL) and Percent Reduction in Area at Fracture (RA) are measures of ductility

Tensile Test

Necking

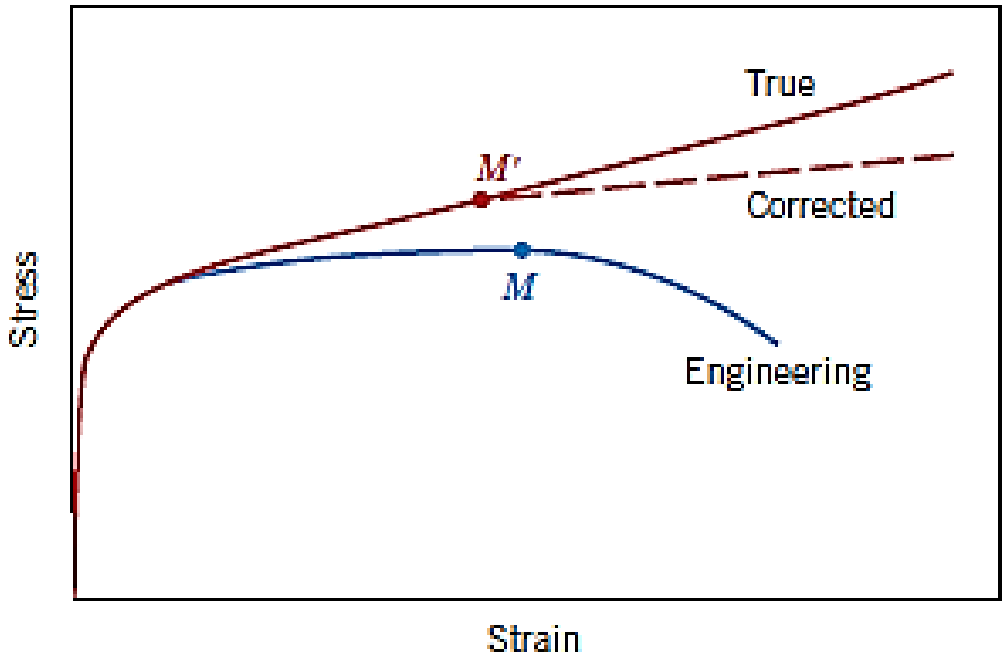


While, for a semicrystalline polymer …



Tensile Test

True Stress and True Strain



$$\sigma = \frac{F}{A_0} \qquad \sigma_T = \frac{F}{A_i}$$

$$\varepsilon = \frac{l_i - l_0}{l_0} \qquad \varepsilon_T = ln\frac{l_i}{l_0}$$

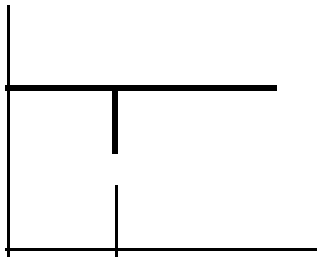
$$\sigma_T = \sigma(1+\varepsilon)$$

$$\varepsilon_T = \ln(1+\varepsilon)$$

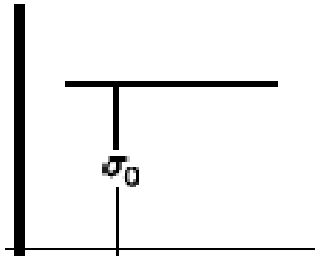
These are valid only up to onset of necking when $A_0 l_0 = A_i l_i$ (constancy of volume)

Tensile Test

Work Hardening/Strain Hardening



Rigid ideal plastic material



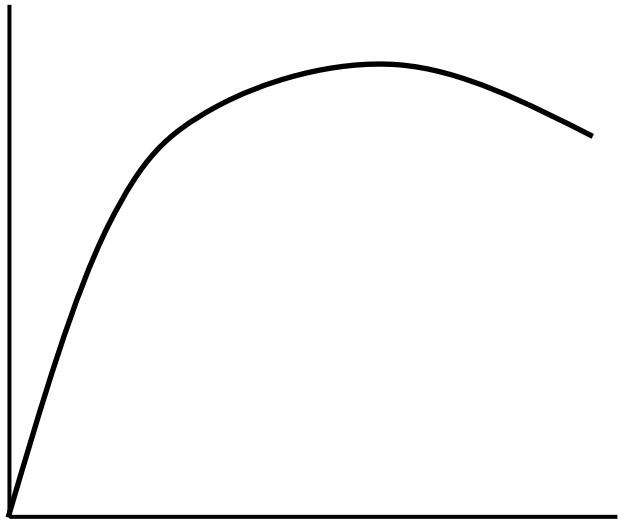
Ideal plastic material with elastic region



Piecewise linear (strain-hardening) material

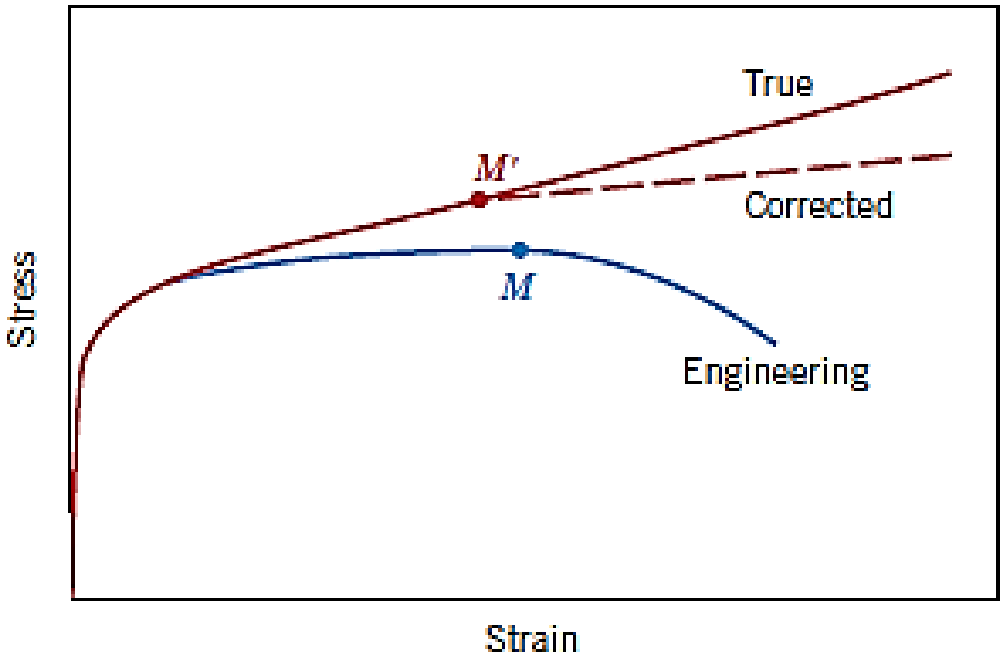
Figure from Mechanical Metallurgy by G.E. Dieter

The stress-strain curve has a positive slope in the plastic region, indicating that the stress required to cause further deformation increases with increasing strain, a phenomenon known as work hardening or strain hardening



Tensile Test

Plastic Flow Curve



Hollomon Equation

$$\sigma_T = K\varepsilon_T^n$$

- Complex stress state within the neck region (i.e., the existence of other stress components in addition to the axial stress)
- Hence, the correct stress (axial) within the neck is slightly lower than the stress computed from the applied load and neck cross-sectional area; this leads to the “corrected” curve
- Strain hardening exponent ‘n’
- Hollomon Eqn. valid from onset of plastic deformation to onset of necking

Tensile Test

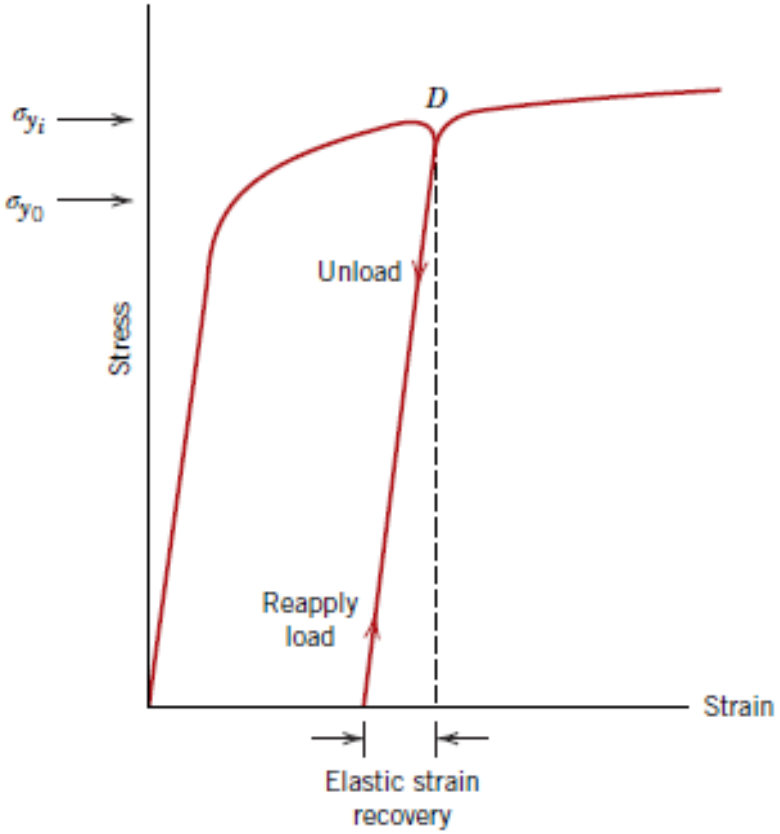
Plastic Flow Curve

Values of n and K for various alloys

		K	
Material	***n***	***MPa***	***psi***
Low-carbon steel (annealed)	0.21	600	87,000
4340 steel alloy (tempered at 315°C)	0.12	2650	385,000
304 stainless steel (annealed)	0.44	1400	205,000
Copper (annealed)	0.44	530	76,500
Naval brass (annealed)	0.21	585	85,000
2024 aluminum alloy (heat-treated—T3)	0.17	780	113,000
AZ-31B magnesium alloy (annealed)	0.16	450	66,000

Tensile Test

Elastic Recovery After Plastic Deformation



- Upon release of the load during the course of a stress–strain test, some fraction of the total deformation is recovered as elastic strain
- The magnitude of this elastic strain, which is regained during unloading, corresponds to the strain recovery
- There will also be an elastic strain recovery associated with fracture